ClimateAdapt Pune: Evidence-Led Climate Action Report

Pilot Study: Pune Urban Core, Maharashtra | Generated: 19 October 2026 04:16 UTC

Executive Summary: This report translates real municipal drainage statistics (OpenCity Pune), OpenStreetMap infrastructure assets (5,779 trees, 267 medical facilities, waterways), Open-Meteo 7-day atmospheric forecasts, and Google Gemini AI into actionable climate resilience priorities for Pune's electoral wards. All findings distinguish measured infrastructure facts from model proxies, with automated MILP budget optimization and expert LLM strategic synthesis.

1. High-Priority Wards (Screening Indicators)

Scores are normalized 0-100 screening proxies. Higher values denote greater exposure or infrastructure deficit.

Ward / Prabhag	Heat Proxy	Flood Proxy	Water Stress	Green Deficit	Composite
Vadgaon Sheri - Ganesh N	58.4	75.2	52.0	95.0	68.8
Arunkumar Vaidya Stadium	64.4	54.8	60.0	95.0	65.0
Ghorpade Garden	63.8	54.6	60.0	93.3	64.4
Upper Indiranagar	58.4	59.1	60.0	95.0	64.4
Mohammadwadi	58.4	60.7	52.0	95.0	63.7
Vimannagar Sanjay Park	58.4	55.4	60.0	95.0	63.1

2. Prioritized Adaptation Action Plan (Budget Allocation)

Target Portfolio Budget: INR 9,000,000 | **Committed Allocation:** INR 840,000 (9.3% efficiency)

Intervention	Category	Units	Cost (INR)	Dept	Lead Time
Cool Roof Reflective Coating	Heat & Urban Co	7 building	■840,000	PMC Building & Dev	1 mo

3. Climate Stress Scenario Comparisons

Scenario	Heat Index	Flood Index	Water Stress	Composite Index
Current Baseline	58.4	51.0	60.0	61.5
Extreme Heatwave (+3.5°C)	78.4	51.0	75.0	70.8
Extreme Monsoon Cloudburst	48.4	76.0	50.0	68.8
Post-Adaptation Plan	50.0	51.0	60.0	58.6

4. Responsible Use & Data Quality Disclosures

- ***Distinction of Evidence:*** *Drainage statistics are verified 2015-2016 OpenCity PMC releases. Atmospheric values represent Open-Meteo multi-variable NWP forecast models and must not be cited as permanent climate normals.*
- ***Explicit Missing Data:*** *Local observation wells (GSDA telemetry) and uncalibrated DEM hydrological runoff grids remain unconfirmed and are strictly excluded from automated physical simulation to prevent false certainty.*
- ***Human Review:*** *Action allocations are exploratory decision-support calculations and require municipal site surveys before civil construction.*

5. AI Ward Resilience Brief: Ward #44 (Gliding Centre)

Analysis Engine: Google models/gemini-3.8-flash | **Zone:** Hadapsar | **Evaluation Date:** 2026-10-19

1. What's Life Like in This Ward?

With zero recorded trees or parks in our ward, walking outside in the 34.3°C afternoon heat feels like standing in front of an open tandoor with nowhere to hide in the shade. During rainy days, even a little shower creates pani bharaav because 90.0% of our roads do not have any rainwater drains at all. Inside homes with tin or concrete roofs, the heat gets trapped till late night, making it very tough for families and children to sleep peacefully.

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2. Problems and What We Can Do About Them

Problem	How Bad Is It?	Why Does This Happen Here?	What Can Be Done?
Street Flooding	Terrible; 90.0% of roads have no drains to carry water away.	Rainwater has nowhere to run, so even light showers pool on the streets.	Build roadside rainwater naalas and clear paths so water can flow away quickly.
Indoor Heat	Very high; temperatures reach up to 34.3°C under blazing sunshine.	Tin sheets and flat cement roofs soak up heat all day and push it inside rooms.	Apply white heat-reflecting paint for rooftops to keep rooms cooler.
No Shade or Green Cover	Severe; the area officially has 0 recorded public trees and 0 parks.	Open land has been paved over without planting roadside neem or peepal trees.	Plant hardy shade trees along wider streets and community corners.
Dirty Water & Waste	Mostly good, but 1.0% of slum homes still lack underground sewer lines.	A few households are disconnected from main sewer pipes, risking dirty runoff.	Connect the remaining 1.0% homes to the underground sewer network.

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3. How Will the Money, Workers and Time Be Used?

Work To Be Done	How Many?	Money (Rs.)	Workers Needed	Time Needed	Who Is Responsible?
White heat-reflecting paint for rooftops	7 building clusters	Rs. 840,000	14 workers	1 month	PMC Building & Development Department

Summary:
• Total money being used:** Rs. 840,000 out of Rs. 9,000,000
• Money left over:** Rs. 8,160,000
• Workers needed:** 14 person-months
• Everything done in:** 1 months

(Note: The remaining Rs. 8,160,000 can be planned in the next round for roadside naalas and tree planting.)

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4. What Should the Ward Officer Do? (Step by Step)

1. **Days 1–5:** Visit the 7 selected building clusters in the crowded pockets. Talk to the residents, explain how the white roof paint works, and get roof access permissions.
2. **Days 6–10:** Buy good quality white heat-reflecting roof paint and cleaning supplies. Bring in the 14 workers and organize them into teams.
3. **Weeks 2–3:** Have the teams sweep, wash, and dry the roofs, then apply the first thick coat of white paint across all 7 building clusters.
4. **Week 4:** Put on the final top coat, inspect the finished roofs with the local residents, and submit the completion report.

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5. How Will Residents Know Things Are Getting Better?

• ■ Homes with white-painted roofs will feel noticeably cooler in the afternoon, bringing relief from the 34.3°C heat.
• ■ Families under treated roofs won't feel heat radiating down from their ceilings late into the night, cutting down the need for noisy fans running at full speed.
• ■ Indoor daytime room temperatures will drop enough that elderly folks and young kids can rest comfortably inside without sweating constantly.